

Climora — Deployment Report

Climora is a full-stack climate disaster preparedness platform. The backend is a Node.js + Express.js REST API deployed on **Render**, the frontend is a React + Vite SPA deployed on **Vercel**, and the database is hosted on **MongoDB Atlas**.

Live URLs

Service	URL
Backend API	https://climora-4aq8.onrender.com
Frontend App	https://climora-omega.vercel.app

Note: Render free-tier services spin down after inactivity. The first request after idle may take ~30 seconds to respond.

1. Backend Deployment (Render)

Platform: Render

Setup Steps

1. Create a Render account at <https://render.com>
2. Click **New** → **Web Service** and connect your GitHub repository
3. Set the **Root Directory** to `backend`
4. Configure the service:
 - **Build Command:** `npm install`
 - **Start Command:** `node server.js`
 - **Environment:** `Node`
5. Add all environment variables listed in Section 4 under the **Environment** tab
6. Click **Deploy**

Health Check

The root path returns a JSON health-check response:

```
GET https://climora-4aq8.onrender.com/
```

```
{
  "message": "Climate Disaster Preparedness API is running 🚀"
}
```

2. Frontend Deployment (Vercel)

Platform: Vercel

Setup Steps

1. Create a Vercel account at <https://vercel.com>
2. Click **Add New** → **Project** and import your GitHub repository
3. Set the **Root Directory** to `frontend`
4. Configure the build settings:
 - **Framework Preset:** `Vite`
 - **Build Command:** `npm run build`
 - **Output Directory:** `dist`
5. Add the frontend environment variables listed in Section 4
6. Click **Deploy**

SPA Rewrite Rule

The `frontend/vercel.json` file contains a rewrite rule to support client-side routing:

```
{
  "rewrites": [{ "source": "/(.*)", "destination": "/index.html" }]
}
```

This ensures all routes (e.g., `/dashboard`, `/login`) are handled by React Router instead of returning a 404.

3. Database Configuration (MongoDB Atlas)

Platform: MongoDB Atlas

Setup Steps

1. Create a MongoDB Atlas account and click **Build a Cluster**

2. Choose a free-tier (Mo) cluster and select a region
3. Under **Database Access**, create a database user with read/write permissions
4. Under **Network Access**, add `0.0.0.0/0` to the IP allowlist (allows connections from Render)
5. Click **Connect** → **Drivers**, copy the connection string, and replace `<password>` with your database user's password
6. Set this connection string as the `MONGO_URI` environment variable on Render — never commit it to source code

The backend will throw a startup error and exit if `MONGO_URI` is missing or invalid.

4. Environment Variables

4.1 Backend Variables

Set these in Render under **Environment** → **Environment Variables**:

Variable	Purpose	Example / Placeholder
PORT	HTTP port the server listens on	5000
MONGO_URI	MongoDB Atlas connection string	mongodb+srv://<user>:<password>@cluster0.example.mongodb.net/
JWT_SECRET	Secret key for signing JWTs	<your-jwt-secret>
JWT_EXPIRES_IN	JWT expiry duration	7d
GOOGLE_CLIENT_ID	Google OAuth 2.0 client ID	<your-google-client-id>
GOOGLE_CLIENT_SECRET	Google OAuth 2.0 client secret	<your-google-client-secret>
WEATHER_API_KEY	OpenWeatherMap One Call 3.0 API key	<your-openweathermap-key>
WEATHER_BASE_URL	OpenWeatherMap base URL	https://api.openweathermap.org/data/3.0/onecall
YOUTUBE_API_KEY	YouTube Data API v3 key (for article video fetch)	<your-youtube-api-key>
CLOUDINARY_CLOUD_NAME	Cloudinary cloud name (for report image uploads)	<your-cloud-name>
CLOUDINARY_API_KEY	Cloudinary API key	<your-cloudinary-api-key>
CLOUDINARY_API_SECRET	Cloudinary API secret	<your-cloudinary-api-secret>
ORS_API_KEY	OpenRouteService key (for shelter routing/distance)	<your-ors-api-key>
NEWSDATA_API_KEY	NewsData.io key (for climate news feed)	<your-newsdata-api-key>
FRONTEND_URL	Deployed frontend origin (used for CORS)	https://climora-omega.vercel.app

4.2 Frontend Variables

Set these in Vercel under **Settings** → **Environment Variables**:

Variable	Purpose	Example / Placeholder
VITE_API_BASE_URL	Backend API base URL used by axios	https://climora-4aq8.onrender.com/api
VITE_GOOGLE_CLIENT_ID	Google OAuth client ID for frontend sign-in button	<your-google-client-id>

5. Deployment Evidence

Backend Health Check

```
GET https://climora-4aq8.onrender.com/
```

Expected response:

```
{
  "message": "Climate Disaster Preparedness API is running 🚀"
}
```

Frontend Live App

The frontend application is live and accessible at: **<https://climora-omega.vercel.app>**

Screenshots

Vercel Production Deployment Dashboard

 Vercel deployment dashboard showing climora-omega.vercel.app with status Ready

Live Frontend Application

 Climora live frontend at climora-omega.vercel.app showing the landing page

If images do not render in your viewer, screenshots are located at `docs/screenshots/` in the repository.

6. Local Development Setup

6.1 Backend

```
# 1. Navigate to the backend directory
cd backend

# 2. Install dependencies
npm install

# 3. Create the environment file
cp .env.example .env
# Then fill in all values from Section 4.1 with real credentials

# 4. Start the development server
npm run dev
```

The backend will be available at **http://localhost:5000**

If any required environment variable is missing, the server will throw a startup error and exit with a non-zero code.

6.2 Frontend

```
# 1. Navigate to the frontend directory
cd frontend

# 2. Install dependencies
npm install

# 3. Create the environment file
echo "VITE_API_BASE_URL=http://localhost:5000/api" > .env
echo "VITE_GOOGLE_CLIENT_ID=<your-google-client-id>" >> .env

# 4. Start the development server
npm run dev
```

The frontend dev server will be available at **http://localhost:5173**